PS2Mamba: A Pyramid-Based Spectral–Spatial Mamba for Hyperspectral Image Classification

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***Abstract*—When hyperspectral image (HSI) classification encounters high-dimensional spectral channels, the feature utilization rate is often low due to significant redundancy between channels and uneven discriminatory power. Furthermore, different land cover types exhibit notable differences in scale and morphological changes across space. This structural diversity poses significant challenges to spatial feature modeling. To address this problem, this article proposes a novel framework based on the Mamba model-PS2Mamba for HSI classification. This framework integrates three strategies: spectral fine modeling, global perceptual scale adaptation, and multiscale spatial structure modeling. First, this article designs a statistically enhanced normalized band refinement (SENBR) module, which dynamically enhances or suppresses channel features based on channel correlation, variability, and importance, effectively suppressing redundant and noisy bands. Second, a global-aware scale adaptation (GASA) module is proposed. By incorporating a scale scoring network and a multidirectional modeling mechanism, this module enables adaptive perception and directional enhancement modeling of spatial structures at different scales. Finally, a lightweight multiscale spatial structure extraction module, LightPyramid, is constructed. By enriching spatial semantic information through multibranch parallel convolutions, it enhances the model's ability to represent complex surface structures, while maintaining spatial resolution. Experimental results on four representative hyperspectral datasets, including Pavia University (PU), Salinas (SA), and two unmanned aerial vehicle (UAV)-based datasets (HongHu and HanChuan), demonstrate that PS2Mamba achieves overall accuracies of 98.77%, 99.61%, 96.20%, and 96.01%, respectively. Compared with existing convolutional neural network (CNN)-, graph convolutional network (GCN)-, Transformer-, and Mamba-based models, PS2Mamba achieves up to 12.91% improvement in accuracy, showing superior generalization and robustness, particularly under small-sample conditions.**

***Index Terms*—Deep learning, hyperspectral image (HSI) classification, Mamba model, small sample.**

I. Introduction

With the continuous development of Earth's observation technologies, hyperspectral images (HSIs) are now more widely available. Unlike traditional imaging systems that capture only the red, green, and blue (RGB) channels, HSIs cover a broad range of wavelengths, from visible light to near infrared, mid infrared, and even far infrared, typically encompassing dozens or even hundreds of contiguous spectral bands. Thanks to its rich spectral information, HSI can identify targets that are hard to distinguish in conventional RGB images, significantly enhancing the accuracy of surface information recognition. Based on these advantages, the hyperspectral imaging has found broad applications in fields, such as biomedicine [1], [2], precision agriculture [3], [4], vegetation analysis [5], and mineral mapping [6]. In these applications, HSI classification plays a pivotal role, aiming to assign category labels to each pixel in the image.

With the continuous advancement of deep learning techniques, HSI classification has achieved remarkable progress in recent years. Current HSI classification methods can be broadly categorized into two types: one based on traditional machine learning algorithms [7], [8] and the other relying on deep learning approaches. The latter has garnered significant attention in recent years and can be further subdivided into various architectural variants, including those based on graph convolutional networks (GCNs) [9], [10], [11], convolutional neural networks (CNNs) [12], [13], [14], [15], and Transformers [16], [17], [18], [19], [20].

Hong et al. [21] proposed SpectralFormer. SpectralFormer proposed a spectral grouping embedding module that can learn richer local spectral representations. Li et al. [22] proposed the channel-layer oriented lightweight network (CLOLN) model. The CLOLN model is a lightweight network oriented toward the channel layer, aimed at reducing the number of parameters and computational complexity associated with CNN. Zhou et al. [23] proposed the attention multihop graph and multiscale convolutional fusion network (AMGCFN). AMGCFN proposes multihop graphs to expand the narrow receptive field of GCN, forming a cascaded network by stacking multihop graphs to effectively extract the structural information. Roy

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et al. [24] proposed MorphFormer, an attention-based model that integrates spectral, spatial, and morphological features. The proposed spectral–spatial–shape attention Transformer enhances the classification accuracy of HSIs while improving the model's robustness to spectral and spatial variations. Sun et al. [17] proposed the spectral–spatial feature tokenization Transformer (SSFTT). SSFTT proposes a Gaussian distribution weighted labeling module that can transform shallow spatial–spectral features into labeled semantic representations. Zhao et al. [25] proposed the groupwise separable convolutional vision transformer (GSC-Vit) model, which combines separable convolution with Vision Transformer (ViT) to capture local and global spectral spatial information for HSIs classification. Mohamed et al. [26] proposed FactoFormer, a factorized hyperspectral Transformer that decouples spectral and spatial representations and adopts self-supervised pretraining to enhance spectral–spatial feature learning. Hong et al. [9] proposed a GCN framework that constructs spectral–spatial graphs and dynamically captures nonlocal structural relations for HSI classification. Ahmad et al. [27] developed a 3D-CNN with active transfer learning, which combines 3-D convolutional feature extraction with iterative sample selection to improve classification under limited labeled data.

Mainstream CNN, Transformer, and GCN models all have certain structural limitations in HSI classification tasks. Due to the inherent local receptive field of CNN models, they struggle to effectively capture long-range dependencies in images, limiting their ability to model global spatial semantics [11], [14], [28], [29]. In contrast, although Transformer models perform better in handling global context, the core self-attention mechanism experiences quadratic growth in computational complexity as the sequence length increases, resulting in significant consumption of computational resources, especially when performing pixel-level fine feature extraction [30], [31], [32], [33]. Meanwhile, GCN models also expose several key issues in HSI classification. First, their performance heavily depends on the way the graph structure is constructed, and manually set adjacency relations may not fully express the complex spectral and spatial features. Second, GCN primarily aggregates local neighborhood information, making it difficult to effectively capture long-range spatial or spectral dependencies [9], [34], [35], [36]. The existence of these issues motivates us to explore a novel HSI classification framework, one that can model remote dependencies while maintaining the linear computational complexity, thus more efficiently addressing the challenges of modeling high-dimensional spatial data [37], [38], [39]. Li et al. [40] proposed the MambaHSI model. MambaHSI is the first image-level HSI classification model based on the Mamba architecture. It efficiently models long-range dependencies across the entire image with linear complexity and adaptively fuses spatial and spectral features, thereby significantly improving the classification performance. Pan et al. [41] introduced the MambaLG model, a dual-branch HSI classification method that combines local and global spatial modeling with spectral awareness. It aims to enhance the global modeling capability while preserving local details, thus achieving higher classification accuracy and efficiency.

Despite the aforementioned advantages of Mamba, there are still several issues that need attention when applying it to HSI classification tasks [42], [43].

1) The global learning mechanism, while effective for capturing global spatial semantic information, tends to overlook local spatial details. Mamba achieves efficient global modeling through 1-D linear scanning, but this requires converting 2-D image data into 1-D sequences. This transformation may disrupt the spatial relationships of adjacent pixels, compromising the integrity of the local spatial structure.
2) HSIs contain hundreds of continuous, overlapping bands, many of which are redundant or noisy. This high dimensionality can lead to the "curse of dimensionality," where data analysis complexity grows exponentially with feature dimensions, hindering efficient processing. High-dimensional features can dilute useful information, affecting the model's discriminative ability, especially in small-sample situations. With limited training samples, the model struggles to learn stable feature distributions, leading to overfitting and poor generalization. Thus, effectively reducing dimensionality while retaining key spectral information and enhancing robustness to limited samples remains a key challenge.

To address the above limitations, this article delves into the characteristics of HIS and proposes a novel PS2Mamba method. The main contributions of this article are summarized as follows.

1) A statistically enhanced normalized band refinement (SENBR) module is carefully designed to address the issues of spectral channel redundancy and uneven channel importance in HSIs. This module introduces three new statistical indicators: channel correlation, channel variability, and channel importance, to dynamically weight the spectral channels without adding extra supervision information. By adaptively strengthening the discriminative spectral bands and suppressing redundant and noisy channels, the module improves the model's efficiency and robustness in representing hyperspectral features.
2) A novel spatial-aware module, global-aware scale adaptation (GASA), is developed. GASA can dynamically select the most suitable feature map scaling factor based on the global context information of the input features, enabling efficient switching and fusion between multiple scales. This enhances the model's robustness in complex scenarios and achieves spatial feature representation with direction consistency and scale adaptability.
3) We introduce LightPyramid, a lightweight multiscale feature enhancement module. Through a multibranch parallel convolutional structure, both local and global spatial semantics are extracted while maintaining the spatial resolution. This significantly enhances the model's ability to represent objects at different scales and can be seamlessly integrated into the backbone network, further improving the overall model's discriminative performance in complex scenarios.

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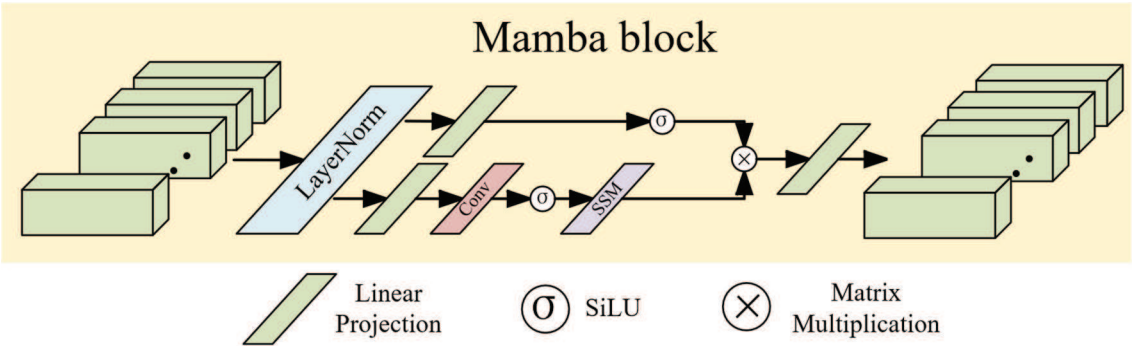


Fig. 1. Structure of Mamba.

4) A compact and high-accuracy HSI classification framework, PS2Mamba, is proposed. It supports full-image-level input and end-to-end prediction, avoiding the spatial structure degradation and information redundancy issues inherent in common patch-based methods. Extensive experiments on four representative datasets (Pavia University (PU), Salinas (SA), HongHu, and HanChuan) demonstrate that PS2Mamba significantly outperforms some state-of-the-art methods in terms of accuracy, robustness, and generalization ability.

The remainder of this article is organized as follows. Section II provides a detailed introduction to the overall structure of PS2Mamba and the detailed process of each module. Section III presents a large number of experimental results and comprehensive analysis, including ablation studies, quantitative comparisons, and visualizations. Section IV summarizes the work presented in this article and outlines potential directions for future research.

II. Methodology

A. Preliminaries

Recently, structured state-space models (SSMs) have gained significant attention in sequence modeling and visual representation tasks due to their ability to efficiently capture long-range dependencies with linear-time complexity. Mamba can be regarded as an advanced extension of the classical state-space model (SSM), and its architecture is illustrated in Fig. 1. Consider an input sequence $x(t) \in R^N$. SSM is formulated as a linear time-invariant dynamical system, which establishes a mapping from the input sequence $x(t)$ to the output sequence $y(t)$. The governing equations of the system are given by

$$h'(t) = Ah(t) + Bx(t) \tag{1}$$

$$y(t) = Ch(t) \tag{2}$$

where (1) describes the state transition dynamics and (2) specifies the output relation. Here, $h(t) \in R^N$ denotes the hidden state vector, and $h'(t)$ represents its temporal derivative. The state evolution is parameterized by the transition matrix $A \in R^{N\times N}$, while $B \in R^{N\times 1}$ and $B \in R^{N\times 1}$ and $C \in R^{N\times 1}$ correspond to the input and output matrices, respectively.

Since SSM is inherently defined in continuous time, a discretization procedure is required to handle discrete-domain data typically encountered in image and signal processing tasks. To this end, the zero-order hold method is employed, transforming the continuous parameters (A, B) into their discrete counterparts $(\bar{A}, \bar{B})$. This process can be expressed as follows:

$$\bar{A} = \exp(\Delta A) \tag{3}$$

$$\bar{B} = (\Delta A)^{-1}\left(\exp(\Delta A) - 1\right) \cdot \Delta B \tag{4}$$

where Δ denotes the sampling interval. Following this discretization, the system dynamics in (1) and (2) can be equivalently reformulated as follows:

$$h_t = \bar{A}h_t + \bar{B}x_t \tag{5}$$

$$y_t = Ch(t). \tag{6}$$

Furthermore, the recursive updates in (5) and (6) admit a parallelized formulation, which greatly improves the computational efficiency. Specifically, the output sequence can be represented as a convolutional operation

$$y = x * \bar{K} \tag{7}$$

$$\bar{K} = \left(C\bar{B}, C\bar{A}\bar{B}, \ldots, C\bar{A}^t\bar{B}\right) \tag{8}$$

where $*$ denotes the discrete convolution and $\bar{K}$ serves as the convolutional kernel constructed from the discretized system parameters.

This discrete convolutional formulation enables parallelized and memory-efficient implementation, providing the theoretical foundation upon which Mamba introduces selective state updates and adaptive parameterization to enhance thr modeling capacity and computational scalability.

B. Overall Structure

The input HIS is represented as $X \in R^{H\times W\times C}$, where $H \times W$ represent the spatial dimensions of the input data, and C represents the number of spectral bands in HSIs. The proposed PS2Mamba mainly consists of three parts: SENBR, GASA, and a lightweight multiscale feature enhancement module (LightPyramid). It is worth noting that our embedding layer maps the spectral vector corresponding to each pixel into the embedding space. Unlike existing image block-based embedding methods, our method performs feature mapping at the pixel level, thus obtaining finer-grained pixel representations. This design is more suitable for dense prediction tasks like hyperspectral classification and is more compatible

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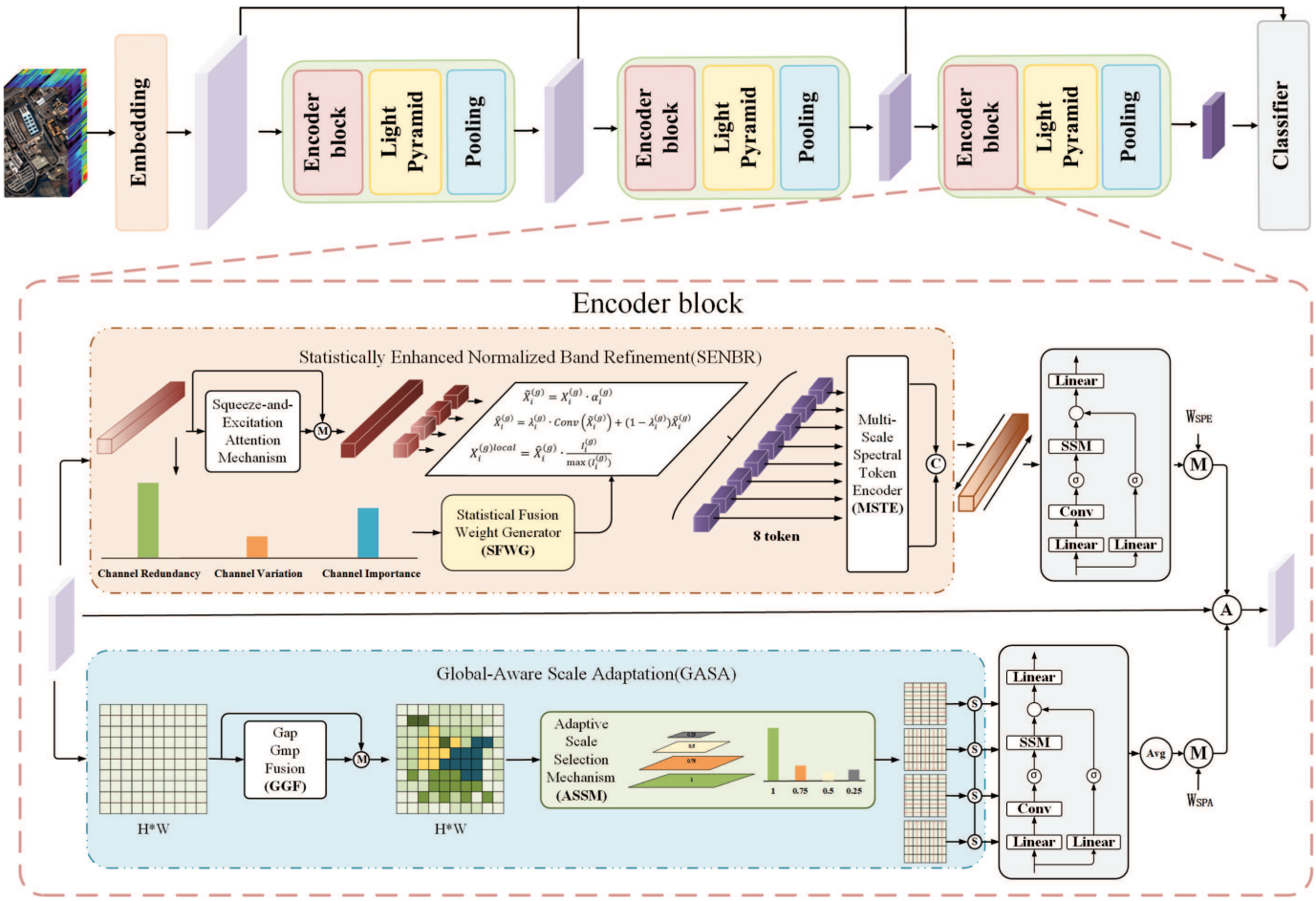


Fig. 2. Overall structural diagram of PS2Mamba.

with models like Mamba, which excel at capturing long-range dependencies in sequence modeling. The designed encoder block is utilized to extract discriminative spatial–spectral features for classification. Spatial and spectral features are crucial for HSI classification, and integrating spatial and spectral information is beneficial for classification.

To effectively integrate complementary information from spatial and spectral domains, an adaptive spectral–spatial integration module (ASSIM) is proposed. Two learnable weights are introduced to adaptively fuse the extracted spatial and spectral features. These weights are optimized via backpropagation, enabling the model to automatically determine the contribution of each modality for robust representation learning. At the same time, we incorporate the concept of residual learning to alleviate potential overfitting during training. The classifier uses a 1×1 kernel convolutional layer to obtain the final prediction result.

The overall framework of the proposed PS2Mamba is shown in Fig. 2.

C. SENBR

To achieve precise modeling of interchannel differences in HSIs, we propose three statistical indicators in this article to measure the structural features of each channel, i.e., channel correlation, channel variability, and channel importance. These weights do not rely on labels or external information and are only calculated from the data in the current input feature map, reflecting the model's adaptability and generality.

Specifically, the input to the model is a feature map of size $[B, C, H, W]$, where B represents the batch size, C represents the number of spectral channels, and C represents the number of spectral channels, and H and W represent the height and width of the image, respectively. When calculating these three types of channel weights, the spatial dimensions are first flattened into a 1-D vector to statistically compute the response distribution of each channel across the entire image. To better capture and differentiate the contributions of each channel, a more detailed characterization of their statistical behavior is necessary. This leads to the definition of three distinct types of channel weights, each highlighting different aspects of the spectral responses. The channel correlation, channel variability, and channel importance are introduced as follows.

Channel Correlation: This measures the degree of information redundancy between channel i and other channels. The covariance matrix between channels is calculated and normalized to obtain the correlation coefficient. Finally, the average correlation coefficient between each channel and the other channels is used as the channel correlation. This indicator can

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be represented as follows:

$$R_i = \frac{1}{N-1} \sum_{j \neq i} \frac{\mathrm{Cov}\left(C_i, C_j\right)}{\sqrt{\mathrm{Cov}\left(C_i, C_i\right) \mathrm{Cov}\left(C_j, C_j\right)}} \tag{9}$$

where R_i represents the correlation for channel i, N is the total number of channels in the HSI, and $\mathrm{Cov}(C_i, C_j)$ is the covariance between channel i and channel j. Equation (9) computes the correlation coefficient for channel i by comparing its response with every other channel in the image. The summation over j represents the averaging of correlations with all other channels, excluding the correlation of i with itself. The covariance terms are normalized to ensure that the correlation is bounded between -1 and 1, where 1 indicates the perfect correlation, and 0 indicates no correlation.

Channel Variability: This measures the degree of variation of each channel in the spectral sequence. The smoothness of the spectral curve is measured using the average absolute difference between adjacent bands. This indicator can be represented as follows:

$$V_i = \frac{1}{L-1} \sum_{k=1}^{L-1} \left| x_{i,k+1} - x_{i,k} \right| \tag{10}$$

where L represents the number of spectral bands and $x_{i,k}$ denotes the response of channel i at band k. A larger V_i indicates that the spectral curve of the channel is not smooth and has abrupt changes. This can be used to dynamically adjust the convolution kernel to smooth high-frequency noise or edges.

Channel Importance: This indicator combines the global variance of the channel with spectral entropy to assess the strength of the discriminative information carried by the channel. To compute the channel importance, the channel response is first normalized to construct a probability distribution. The channel importance for the channel can be represented by the following equation:

$$I_i = \mathrm{Var}\left(C_i\right) \times \left(- \sum_k p_{i,k} \log_2 \left(p_{i,k}\right) \right) \tag{11}$$

$$p_{i,k} = \frac{x_{i,k}}{\sum_k x_{i,k}} \tag{12}$$

where $\mathrm{Var}(C_i)$ represents the variance of channel i and $p_{i,k}$ is the probability distribution of the response of channel i at band k. The probability distribution is defined as (12). This formulation evaluates the channel's global variance and entropy to determine the degree of discriminative information contained in the channel. High variance and low entropy indicate that the channel carries strong and focused discriminative power, whereas low variance and high entropy suggest that the channel contains less useful information.

The above three statistical indicators, respectively, constitute the guiding basis for redundancy suppression, smoothing control, and information enhancement in the channel dimension. This statistical-driven mechanism not only does not rely on label information but also automatically adapts to the current input feature distribution during each forward propagation, thereby enhancing the model's dynamic modeling ability for spectral features in different scenarios.

The statistical fusion weight generator (SFWG) module calculates channel statistical indicators on a global scale, constructs correlation weights, smoothing weights, and importance weights, and constructs a unified global weight term

$$\alpha_i = 1 - R_i \tag{13}$$

$$\lambda_i = \frac{v_i - \min\left(V\right)}{\max\left(V\right) - \min\left(V\right)} \tag{14}$$

$$\gamma_i = \frac{I_i}{\max\left(I\right)} \tag{15}$$

$$\omega_i^{\mathrm{global}} = \alpha_i \cdot \lambda_i \cdot \gamma_i. \tag{16}$$

Equation (13) represents the inverse redundancy ratio between a channel and other channels. High correlation will result in weights approaching 0, thereby suppressing the response of the channel; low correlation leads to weights approaching 1, thereby preserving channel responses with significant information differences and enhancing information diversity. Equation (14) is used as the normalized spectral variability index to control the intensity of spectral convolution smoothing. Its design ensures that strong smoothing is applied to high variance bands to suppress noise, while low variance bands retain the original detailed features, thus achieving a good balance between local smoothing and structural preservation. Equation (15) reflects the relative discriminative power of channels. The higher the importance, the more significant the response enhancement, allowing the model to focus on key bands in the spectrum with higher information density and clearer category distribution. This normalization strategy ensures that the weight distribution is within a controllable range, avoiding extreme amplification or suppression. Achieving a unified control strategy for local structural perception and global information alignment in spectral modeling involves introducing a mechanism that combines group-based enhancement and global fusion in the SENBR module. This mechanism consists of two stages. First, channel features within each group are finely modeled based on correlation, variability, and information strength. Second, the global statistical weights mentioned earlier are introduced across the entire channel range to uniformly adjust the output of each group, realizing local–global feature fusion. The group modeling and local enhancement mechanism are illustrated as follows.

Correlation Enhancement Mechanism: This mechanism is based on the channel correlation coefficient R_i defined earlier and suppresses redundant channels within the group. Specifically, for the feature response $X_i^{(g)}$ of each group, the independence weight is constructed. This weight is then utilized for the channel enhancement operation

$$\tilde{X}_i^{(g)} = X_i^{(g)} \cdot \alpha_i^{(g)}. \tag{17}$$

Variability Enhancement Mechanism: For the feature response $X_i^{(g)}$ of each group, the previously introduced smoothing weight is constructed. Subsequently, a fixed-size 1-D convolutional kernel is applied to perform sliding convolutional smoothing along the spectral dimension. The convolutional kernel is learned during training, and its output

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is weighted by the smoothing weight to control the smoothing amplitude

$$\hat{X}_i^{(g)} = \lambda_i^{(g)} \cdot \mathrm{Conv}\left(\tilde{X}_i^{(g)}\right) + \left(1 - \lambda_i^{(g)}\right) \tilde{X}_i^{(g)}. \tag{18}$$

This operation fuses the original features of the current channel with the average information from its spectral neighborhood, and the smoothing intensity is adaptively adjusted by spectral variability. For bands with high variability and high variance, the corresponding $\lambda_i^{(g)}$ is higher, and the model output relies more on the smoothing convolution, whereas for channels with low variability and stable structure, more of the original response information is retained. This mechanism can significantly reduce the risk of misjudgment caused by high-frequency disturbances, while suppressing ineffective noise, thus maintaining the structure of the spectral curve. It provides more stable foundational features for subsequent discriminative enhancement and sequence modeling.

Importance Enhancement Mechanism: For the feature response $X_i^{(g)}$ of each group, the importance of the weight mentioned earlier is constructed. After normalization, it is adopted for the weighted enhancement of the feature response

$$X_i^{(g)\mathrm{local}} = \hat{X}_i^{(g)} \cdot \frac{I_i^{(g)}}{\max\left(I_i^{(g)}\right)}. \tag{19}$$

This approach ensures that channels with high information density and strong responses are further highlighted, effectively improving the model's discriminative ability for key bands. Finally, by merging local enhancement with global weights, unified spectral regulations are achieved

$$X_i^{(g)\mathrm{final}} = X_i^{(g)\mathrm{local}} \cdot \omega_i^{\mathrm{global}}. \tag{20}$$

This "local modeling followed by global fusion" approach not only enhances the model's sensitivity to local spectral structural features but also maintains the semantic consistency of channel importance at the global scale. It provides high-quality input features that are structurally clear and globally aligned for the subsequent Mamba module, forming a good complement to its long-range dependency modeling mechanism. This significantly improves the discriminative ability and expression stability of spectral features. The details of the "local modeling followed by global fusion" approach are provided in Algorithm 1.

To further model the spectral local context, the model divides the output of each group into eight subgroups and feeds them into the multiscale spectral token encoder (MSTE) module. Specifically, multiple convolutional kernels with different receptive fields $k \in \{1, 3, 5\}$ are applied to slide along the spectral dimension to extract band structure features at different scales. For each subgroup S_j, the multiscale convolution output is

$$F_j = \mathrm{Concat}\left(\mathrm{Conv}_1\left(S_j\right), \mathrm{Conv}_3\left(S_j\right), \mathrm{Conv}_5\left(S_j\right)\right). \tag{21}$$

After all scale branch outputs are concatenated along the channel dimension, a 1×1 convolution is applied to fuse them, unify the feature dimensions, and compress information redundancy

$$F_j^{\mathrm{fused}} = \mathrm{Conv}_1\left(F_j\right). \tag{22}$$

Algorithm 1 Local Structural Perception and Global Information Alignment in Spectral Modeling

Input: Feature map $X \in \mathbb{R}^{B\times C\times H\times W}$,
Output: Final enhanced feature map X_{final}

```
Pass X through SE channel attention mechanism → X_processed
Group the processed features into G subgroups: X_grouped
for each group g = 1 to G do
    R_g ← Compute − Correlation(X_grouped[g])
    V_g ← Compute − Variability(X_grouped[g])
    I_g ← Compute − Importance(X_grouped[g])
    Apply correlation enhancement → X_grouped[g] ← X_grouped[g] × ComputeCorrelationWeight(R_g)
    Apply variability enhancement → X_grouped[g] ← Conv − Operation(X_grouped[g], ComputeVariabilityWeight(V_g))
    Apply importane enhancement → X_grouped[g] ← X_grouped[g] × Normalized(I_g)
end for
```

Subsequently, the outputs of all subgroups are concatenated, and a bidirectional spectral scanning mechanism is applied. A sliding window extracts context dependency information along both the forward and reverse band sequences. This process is equivalent to constructing a shallow bidirectional recurrent structure or gated sliding convolution along the channel dimension to capture the potential structural dependencies between distant bands. Finally, the concatenated spectral feature sequence is fed into the lightweight sequence modeler, Mamba. This module efficiently models long-range dependencies across channels using state-space modeling capabilities, without relying on traditional self-attention mechanisms

$$Z = \mathrm{Mamba}\left(F^{\mathrm{scanned}}\right). \tag{23}$$

The output sequence Z serves as the final feature representation for the SENBR module.

D. GASA

To explore the rich coupling between spatial and spectral information in HSIs, we propose a modular framework for pixel-level classification. The framework takes the entire spatial context of the image as input, guiding the model through a process that includes global spatial perception, scale-adaptive selection, and directional enhancement to obtain highly discriminative spatial feature representations. The model starts with the GapGmpFusion(GGF) module, which performs global spatial modeling by integrating global average pooling (GAP) and global max pooling (GMP). These operations capture significant regions and background distributions at the image level, creating spatial features that possess semantic global perception. The structure of the GGF module is shown in Fig. 3.

Next, the model employs the GASA module, which further adapts the structural features at multiple scales. The most appropriate scale is selected dynamically through an adaptive scoring mechanism, and directional enhancement strategies are

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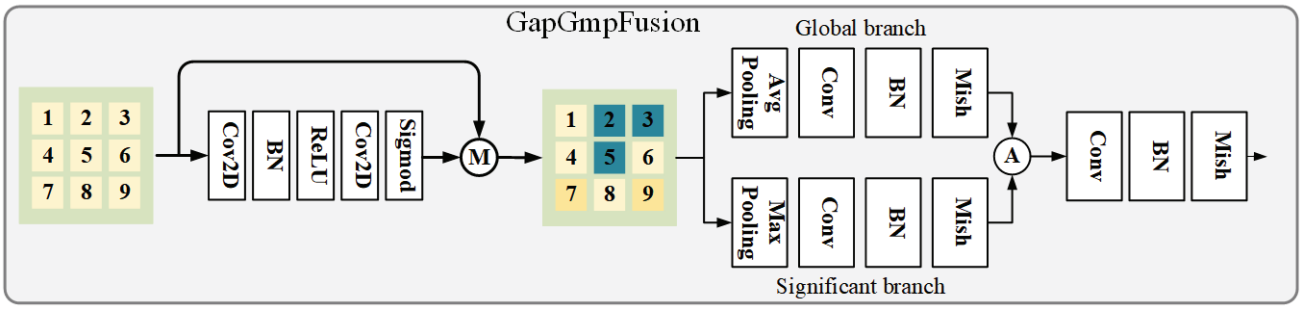


Fig. 3. Structure of GGF.

applied to model features from different spatial transformation perspectives, enhancing the model's ability to perceive shapes, edges, and structural directions. Finally, the fused spatial features are sent to the classification head or the spectral modeling module for pixel-level classification.

Experiments demonstrate that incorporating these spatial modeling strategies significantly improves the model's adaptability to complex object structures and scale variations, achieving excellent classification performance on multiple standard hyperspectral datasets.

To improve the model's overall perception of the spatial structure and provide richer spatial features for subsequent scale selection, the GGF module is introduced. This module, using a lightweight spatial attention mechanism, highlights structural edges and significant regions in the image. The enhanced spatial information is processed through GAP and GMP, which capture both background trends and key region responses. The outputs of these operations are then passed through feature transformation units to increase discriminative power. The resulting fused features are forwarded to the adaptive scale selection mechanism (ASSM), enhancing the model's ability to handle different spatial structures, especially when dealing with classes that have similar spectral characteristics but differ in spatial distribution.

To address the limitations of fixed-scale feature extraction, the ASSM mechanism enables the model to adaptively select the most suitable feature scale from a predefined set. This selection is based on global contextual information, allowing the model to handle both large-scale structures and fine-grained details effectively. The scale selection process, which uses a lightweight scale scoring network, is guided by adaptive scores that allow for flexible and efficient spatial modeling. During training, the Gumbel-softmax sampling mechanism is used to optimize the scale selection process, and in the inference phase, the scale with the highest score is chosen for processing. This multiscale selection significantly enhances the model's adaptability and reduces computational complexity, striking a balance between efficiency and performance. The details of the ASSM mechanism are provided in Algorithm 2.

Algorithm 2 ASSM

Input: Feature map $X \in \mathbb{R}^{B\times C\times H\times W}$
Parameters: Scale set $S = \{1.0, 0.75, 0.5, 0.25\}$ and τ
Compute global average pooling on X along spatial dimensions $\rightarrow x_{avg} \in \mathbb{R}^{B\times C}$
Apply two-layer MLP:
  $x_{fc1} = \text{ReLU}(\text{Linear}(x_{avg}))$
  $x_{\text{logits}} = Linear(x_{fc1}) \in \mathbb{R}^{B\times|S|}$
Apply LogSoftmax to logits $\rightarrow s_{\text{logprob}}$
if training **then**
  Sample $s_{weight} = \text{GumbelSoftmax}(s_{\text{logprob}}, \tau)$
  Update temperature: $\tau \leftarrow \tau \times 0.95$
else
  Set $s_{weight} = OneHot(\arg\max(s_{\text{logprob}}))$
end if
Determine selected scale index $k = \arg\max(s_{weight})$
Set selected scale factor $r = S[k]$
if $r < 1.0$ **then**
Downsample X via bilinear interpolation to size
else
  Keep X unchanged
end if
Output: Rescaled feature map $X_{scaled} \in \mathbb{R}^{B\times C\times rH\times rW}$

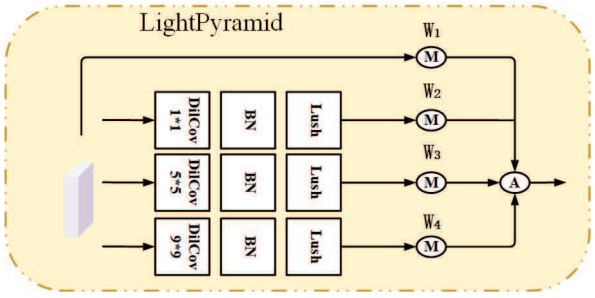


Fig. 4. Structural diagram of LightPyramid.

E. LightPyramid

To enhance the model's ability to perceive structures at different spatial scales in HSIs, this article proposes a novel multiscale feature fusion structure—LightPyramid module. The core idea is to introduce multireceptive field enhancement while controlling the overall computational complexity, thus improving the model's robustness and expression diversity during the spatial modeling process. The LightPyramid module, as shown in Fig. 4, consists of three parallel paths with consistent structure but different receptive fields, corresponding to convolutional perception at different scales.

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Assume specifically, the input features are first passed through three dilated convolutional branches and one identity mapping branch. Among these, the three branches use dilated convolutions with kernel sizes of $(1 \times 1, 5 \times 5, 9 \times 9)$ to capture spatial structural features at different scales: local, small range, medium range, and large range. These dilated convolutions are implemented using depthwise separable convolutions to effectively reduce the parameter overhead. Each branch consists of a dilated convolutional layer, a batch normalization layer, and a Lush activation function. Here, we use the Lush function that we proposed earlier [43].

Lush activation function is a novel nonlinear activation function designed to address the limitations of traditional activation functions like rectified linear unit (ReLU) and Sigmoid, particularly in terms of gradient flow and model expressiveness. The function is defined as follows:

$$\text{Lush}(x) = \begin{cases} x \cdot \tanh\left(\log\left(1 + e^{x}\right)\right), & x < 0 \\ x, & 0 \le x. \end{cases} \tag{24}$$

For negative input values, the Lush activation function applies a smooth nonlinearity through the combination of softplus and tanh, while for positive input values, it behaves like the ReLU activation, returning the input directly. In this way, Lush improves gradient propagation and provides stronger learning capabilities in deep networks. In the LightPyramid module, the Lush activation function enhances the representational power of each branch, allowing the network to learn more complex spatial patterns without significantly increasing computational overhead.

In addition to the three convolutional paths, the original features are also passed through an identity mapping path as the base input, preserving the unaltered features in the residual structure, thereby enhancing the stability and information integrity of the training process.

Each branch's output is multiplied by learnable channel fusion weights W_1–W_4, which represent the importance of the different scale features in the current task. These weights are automatically learned through softmax regularization during training, ensuring that the fused scale features have dynamic adjustment capabilities. Finally, the multiscale information is integrated through feature addition, resulting in a unified dimensional feature map.

By combining the LightPyramid and Mamba modules, the overall model architecture achieves excellent complementarity. The LightPyramid module enhances the local spatial multiscale structure, while the Mamba module captures long-range dependencies in sequence modeling. Together, they significantly improve the model's ability to represent complex multiscale, multiobject scenes in HSIs, all without a significant increase in computational overhead.

This synergy ensures that the model effectively utilizes both local and global spatial information, making it highly suited for analyzing complex HSIs without an undue rise in computational complexity.

III. Experimental Results and Analysis

To thoroughly validate the performance of the proposed PS2Mamba, four typical hyperspectral datasets were used for testing, including PU dataset, SA dataset, and two unmanned aerial vehicle (UAV) datasets from Wuhan University: HanChuan and HongHu. These four datasets exhibit significant differences in scene types and spectral characteristics, which helps to assess the model's generalization ability and classification performance from multiple dimensions. We will provide a detailed description of each dataset.

A. Dataset Description

PU: Acquired by the ROSIS sensor over Pavia, Italy, the dataset has a spatial resolution of 1.3 m, with dimensions of 610×340 pixels and 103 spectral bands. It covers nine land cover categories, such as "trees," "bare soil," and "asphalt." The target classes and sample division are shown in Fig. 5.

SA: Acquired by the AVIRIS sensor in Salinas Valley, CA, USA, this dataset has a size of 512×217 pixels, a resolution of 3.7 m, and 204 spectral bands. It defines 16 land cover classes, including crops like "grape," "lettuce," and "vineyard." The target classes and sample division are shown in Fig. 6.

HongHu: Collected using the Headwall Nano-Hyperspec imager on a DJI Matrice 600 Pro drone in Honghu, Hubei, China, this dataset contains 940×475 pixel images, with a spectral range of 400–1000 nm and 270 spectral bands. It features various crop types and their varieties. The target classes and sample division are shown in Fig. 7.

HanChuan: Acquired by Wuhan University on June 17, 2016, from 17:57 to 18:46 in Hanchuan, Hubei, was captured using a Headwall Nano-Hyperspec hyperspectral camera (17-mm focal length) mounted on a Leica Aibot X6 V1 drone. The dataset covers the urban–rural fringe, including buildings, water bodies, and farmland, with 16 crop classes such as strawberry, cowpea, and soybean. It consists of 1217×303 pixel images with 274 spectral bands spanning 400–1000 nm. The target land cover classes and dataset sample division are shown in Fig. 8.

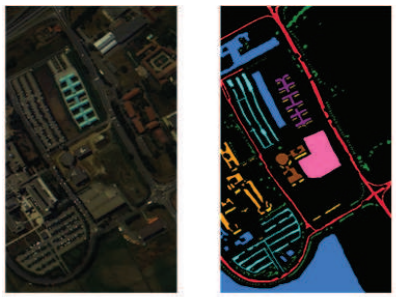
B. Experimental Setup

All experiments in this article were conducted on a platform equipped with an Intel[1] Core[2] i9-14900KF CPU, NVIDIA GeForce RTX 4090D GPU, and 128 GB of random access memory, using the PyTorch framework. In addition, three common evaluation metrics were used to assess the model's classification performance: overall accuracy (OA), average accuracy (AA), and kappa coefficient. The OA value represents the ratio of correctly classified samples to the total number of samples, the AA value represents the average classification accuracy of each category, and the Kappa coefficient is an evaluation metric used to measure the robustness of the model.

The key hyperparameters in our experiments were set as follows: initial learning rate of $3e^{-3}$, the Adam optimizer was employed. The model was trained for 200 epochs with an early stopping patience of 50. For the proposed modules, the group number and token number in SENBR were set to 4 and 16, respectively, and the dilated rates in LightPyramid

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[2]Trademarked.

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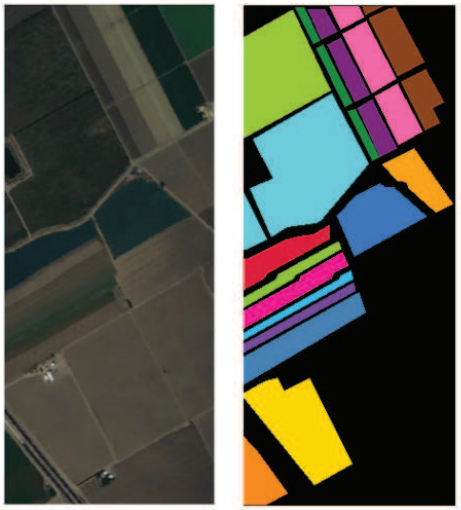


(a) (b)

No.	Color	Class name	Train	validation	Test	Total
1		Asphalt	30	10	6591	6631
2		Meadows	30	10	18609	18649
3		Gravel	30	10	2059	2099
4		Trees	30	10	3024	3064
5		Metal sheets	30	10	1305	1345
6		Bare soil	30	10	4989	5029
7		Bitumen	30	10	1290	1330
8		Bricks	30	10	3642	3682
9		Shadows	30	10	907	947
Total			270	90	42416	42776

(c)

Fig. 5. PU dataset. (a) False color image. (b) Ground truth. (c) Category and sample settings.

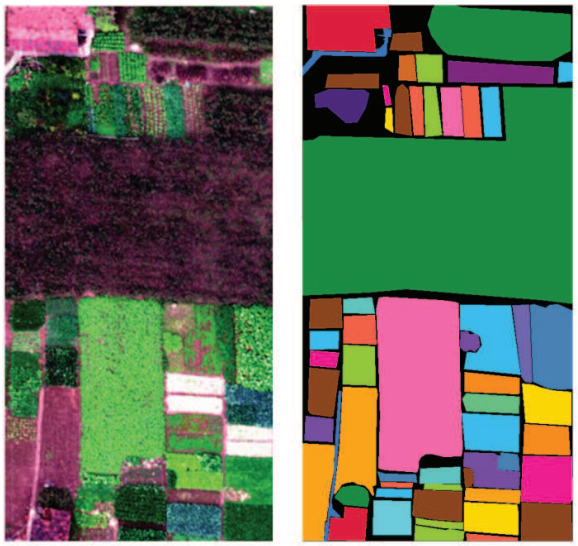


(a) (b)

No.	Color	Class name	Train	validation	Test	Total
1		Brocoli green weeds 1	30	10	1969	2009
2		Brocoli green weeds 2	30	10	3686	3726
3		Fallow	30	10	1936	1976
4		Fallow rough plow	30	10	1354	1394
5		Fallow smooth	30	10	2638	2678
6		Stubble	30	10	3919	3959
7		Celery	30	10	3539	3579
8		Grapes untrained	30	10	11231	11271
9		Soil vinyard develop	30	10	6163	6203
10		Corn senesced green weeds	30	10	3238	3278
11		Lettuce romaine 4wk	30	10	1028	1068
12		Lettuce romaine 5wk	30	10	1887	1927
13		Lettuce romaine 6wk	30	10	876	916
14		Lettuce romaine 7wk	30	10	1030	1070
15		Vinyard untrained	30	10	7228	7268
16		Vinyard vertical trellis	30	10	1767	1807
Total			480	160	53489	54129

(c)

Fig. 6. SA dataset. (a) False color image. (b) Ground truth. (c) Category and sample settings.



(a) (b)

No.	Color	Class name	Train	validation	Test	Total
1		Red roof	30	10	14001	14041
2		Road	30	10	3472	3512
3		Bare soil	30	10	21781	21821
4		Cotton	30	10	163245	163285
5		Cotton firewood	30	10	6178	6218
6		Rape	30	10	44517	44557
7		Chinese cabbage	30	10	24063	24103
8		Pakchoi	30	10	4014	4054
9		Cabbage	30	10	10779	10819
10		Tuber mustard	30	10	12354	12394
11		Brassica parachinensis	30	10	10975	11015
12		Brassica chinensis	30	10	8914	8954
13		Small Brassica chinensis	30	10	22467	22507
14		Lactuca sativa	30	10	7316	7356
15		Celtuce	30	10	962	1002
16		Film covered lettuce	30	10	7222	7262
17		Romaine lettuce	30	10	2970	3010
18		Carrot	30	10	3177	3217
19		White radish	30	10	8672	8712
20		Garlic sprout	30	10	3446	3486
21		Broad bean	30	10	1288	1328
22		Tree	30	10	4000	4040
Total			660	220	385813	386693

(c)

Fig. 7. HongHu dataset. (a) False color image. (b) Ground truth. (c) Category and sample settings.

were [1, 5, 9]. The rationale behind the selection of these values, including sensitivity and hyperparameter analysis, will be provided in Sections III-D and III-F to maintain a concise narrative here.

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(a)



(b)

No.	Color	Class name	Train	validation	Test	Total
1		Strawberry	30	10	44695	44735
2		Cowpea	30	10	22713	22753
3		Soybean	30	10	10247	10287
4		Sorghum	30	10	5313	5353
5		Water spinach	30	10	1160	1200
6		Watermelon	30	10	4493	4533
7		Greens	30	10	5863	5903
8		Trees	30	10	17938	17978
9		Grass	30	10	9429	9469
10		Red roof	30	10	10476	10516
11		Gray roof	30	10	16871	16911
12		Plastic	30	10	3639	3679
13		Bare soil	30	10	9076	9116
14		Road	30	10	18520	18560
15		Bright object	30	10	1096	1136
16		Water	30	10	75361	75401
Total			480	160	256890	257530

(c)

Fig. 8. HanChuan dataset. (a) False color image. (b) Ground truth. (c) Category and sample settings.

To validate the effectiveness of the proposed PS2Mamba method, several advanced networks based on CNNs, Transformers, and GCNs were selected for comparison, including SpectralFormer, CLOLN, AMGCFN, moreFormer, SSFTT, GSC-Vit, mambaLG, and MambaHSI.

SpectralFormer [21]: It introduces a spectral grouping embedding module to learn richer local spectral representations, enhancing spectral feature modeling capability.

CLOLN [22]: It designs a lightweight channel-oriented structure to reduce the parameters and computation in CNNs, balancing efficiency and performance.

AMGCFN [23]: This method effectively extends the GCN receptive field by introducing multihop graphs and multiscale convolutional fusion, enhancing structural information extraction ability.

MorphFormer [24]: An attention-based model that integrates spectral, spatial, and morphological features. The proposed spectral–spatial–shape attention Transformer enhances the classification accuracy of HSIs, while improving the model's robustness to spectral and spatial variations.

SSFTT [17]: It proposes a Gaussian-weighted tokenization mechanism to extract shallow spectral–spatial semantic features from a sequence perspective, achieving efficient Transformer modeling.

GSC-Vit [25]: This method combines grouped separable convolutions with ViT to unify the modeling of local and global spectral–spatial information.

MambaHSI [40]: The first image-level Mamba model that adaptively fuses spatial and spectral features and models long-range dependencies with the linear complexity.

mambaLG [41]: It constructs a dual-branch structure that integrates local spatial modeling with long-range spectral perception, balancing fine feature retention with global modeling.

C. Quantitative Results and Analysis

Tables I–IV present the classification results of all methods on PU, SA, Honghu, and Hanchuan datasets, including OA, AA, kappa coefficient, and the classification accuracy for each category. The best results are highlighted in bold. From Tables II to V, it can be observed that traditional methods based on CNN and Transformer perform poorly when the number of samples is limited, with their classification accuracy being significantly lower than the PS2Mamba method proposed in this article. This is mainly because these methods typically rely on a large number of training samples, and their generalization ability is significantly limited in HSI small-sample scenarios. In contrast, PS2Mamba maintains excellent classification performance under small-sample conditions, validating its strong robustness and effectiveness in HSI classification task.

D. Ablation Experiments

To validate the effectiveness of each submodule in the proposed PS2Mamba model, we conduct ablation experiments on four typical hyperspectral datasets, with the specific results shown in Table V. The PS2Mamba model comprises three key components: SENBR, GASA, and LightPyramid. We systematically evaluate the contribution of each module by testing different configurations and comparing the OA of the model across the datasets.

Seven model combinations were constructed, and the corresponding module configurations and classification accuracies are presented in Table V. In the first case, the model was based on the basic spectral structure without any enhancement modules, resulting in relatively low classification accuracy across all datasets. The introduction of the SENBR module in case 2 led to a significant performance improvement, particularly on the HongHu and HanChuan datasets, where accuracy increased by 7.35% and 9.81%, respectively. This confirms that SENBR, with its statistically-driven channel recalibration mechanism, effectively enhances the discriminative power and robustness of spectral features.

In cases 3 and 4, we evaluate the independent contribution of the GASA module. Case 3, which included only the spatial branch baseline, showed lower performance than case 4, which incorporated the GASA module alone. This result emphasizes the importance of GASA's global scale selection and directional enhancement mechanisms in spatial structure modeling. Notably, on the PU dataset, the addition of GASA resulted in a 6.29% improvement, demonstrating its ability to effectively model spatial structure variations. Cases 5 and 6 were designed to test the combined effect of SENBR and GASA. Case 5 incorporated the spatial–spectral

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TABLE I
QUANTITATIVE RESULT (ACC% ± STD%) OF PU DATASET

Methods	spectralFormer TGRS 2021	CLOLN TGRS2024	AMGCFN TGRS2023	morphFormer TGRS2023	SSFTT TGRS2022	GSCVIT TGRS2024	mambahsi TGRS2024	mambaLG TGRS2025	PS2Mamba Ours
1	85.92+-2.36	**98.27±1.00**	95.81±1.78	95.72±0.79	90.27±1.28	97.10±1.64	94.35±1.73	93.96±2.36	96.63±2.53
2	92.97±2.39	98.98±0.76	98.86±0.64	95.46±1.62	94.58±4.84	97.68±1.32	96.43±1.73	93.57±0.94	**99.69±0.26**
3	84.73±5.17	87.83±7.54	**99.90±0.33**	93.52±0.34	90.41±4.20	86.46±7.42	93.39±5.35	89.95±1.55	98.29±2.66
4	98.90±5.24	**97.80±1.46**	92.08±0.96	97.13±2.49	91.67±±1.36	91.83±9.94	89.75±3.60	97.10±1.17	96.65± 0.35
5	99.92±1.88	99.68±0.59	99.69±5.81	99.84±0.87	99.71±0.20	99.45±0.53	99.00±0.02	99.99±0.01	**100±0**
6	88.40±9.24	85.27±7.50	99.99±0.01	89.95±0.91	95.99±2.07	79.65±9.24	98.60±0.93	99.28±0.18	**100±0**
7	93.85±2.06	85.92±7.00	99.84±0.61	99.53±0.61	98.26±1.06	86.63±10.10	96.50±3.13	99.25±0.75	**99.95±0.06**
8	95.20±5.28	90.93±4.12	94.98±0.76	93.04±0.69	88.33±3.29	89.50±4.70	94.40±2.93	83.84±2.82	**98.4±0.99**
9	98.98±0.71	97.12±2.03	96.72±0.84	99.12±0.97	96.31±2.05	98.71±1.10	99.36±0.62	99.89±0.94	**99.52±0.83**
OA(%)	91.89±3.25	94.97±1.43	97.31±0.74	95.08±1.53	93.17±2.41	91.1±2.60	95.74±0.90	94.07±0.38	**98.77±0.45**
AA(%)	93.21±1.07	93.53±1.59	98.34±0.77	95.29±1.49	93.51±0.68	79.93±2.67	95.86±1.11	95.28±0.76	**98.66±0.33**
Kappa×100	89.24±3.19	93.38±1.84	96.46±4.99	93.40±0.72	90.87±2.59	77.40±4.41	90.38±2.95	92.23±3.62	**99.25±1.5**

TABLE II
QUANTITATIVE RESULT (ACC% ± STD%) OF SA DATASET

Methods	spectralFormer TGRS 2021	CLOLN TGRS2024	AMGCFN TGRS2023	morphFormer TGRS2023	SSFTT TGRS2022	GSCVIT TGRS2024	mambahsi TGRS2024	mambaLG TGRS2025	PS2Mamba Ours
1	97.42±1.57	99.98±2.54	**100±0**	98.43±2.35	99.95±0.11	99.91±0.08	99.93±0.18	100±0	99.99±0.02
2	98.82±0.62	100±0.02	**100±0**	99.16±1.25	99.88±0.28	100±0.01	100±0.01	99.92±1.28	**100±0**
3	97.43±1.48	96.51±2.35	**100±0**	99.94±0.86	100±0	99.93±0.18	99.92±0.15	99.94±0.14	99.90±0.26
4	99.50±0.15	99.64±1.31	96.99±0.41	99.88±0.04	99.89±0.07	**99.95±0.11**	99.90±0.09	99.25±0.58	99.91±0.07
5	95.77±2.38	98.97±0.06	**99.01±0.81**	96.90±0.33	97.95±1.7	98.20±0.53	98.25±0.92	98.83±0.66	98.89±0.63
6	98.98±0.39	99.98±0.02	99.89±0.81	100±0.01	99.96±0.06	99.99±0.01	99.99±0.03	99.99±0.02	**100±0**
7	99.43±0.11	99.70±1.23	99.63±0.35	96.84±0.41	99.95±0.11	99.8±0.29	99.80±0.24	99.97±0.03	**99.90±0.13**
8	71.16±7.84	97.55±0.24	97.54±4.04	85.98±1.03	90.41±3.35	97.28±1.72	96.08±2.35	87.78±1.25	**98.30±1.42**
9	97.33±1.29	99.96±1.05	99.91±0.12	99.93±0.16	99.92±0.14	99.97±1.65	99.97±0.07	99.96±0.02	**99.98±0.04**
10	88.12±5.60	97.60±0.58	**99.01±4.77**	96.45±0.96	98.03±1.23	97.69±0.38	97.55±1.54	96.34±0.29	98.91±0.47
11	96.88±0.89	99.68±0.52	98.71±0.78	99.32±0.56	99.96±0.08	99.71±0.37	**99.91±0.09**	99.33±0.92	99.79±0.45
12	99.05±0.99	97.25±0.28	99.91±1.45	99.98±0.07	99.88±0.20	99.87±0.09	**99.99±0.03**	99.99±0.04	99.94±0.09
13	99.19±0.34	99.24±1.00	99.88±0.71	99.96±0.02	99.65±0.57	99.88±0.08	99.75±0.51	99.97±0.03	**99.98±0.07**
14	95.37±3.03	99.12±0.67	99.42±3.07	97.40±0.32	95.35±7.32	99.89±0.25	99.88±0.20	**100±0**	99.75±0.28
15	67.02±5.92	98.88±2.01	94.63±0.79	86.81±0.95	92.31±4.17	98.23±1.24	97.76±1.85	94.14±0.72	**98.94±1.52**
16	95.81±2.30	99.01±0.24	99.95±0.47	99.32±0.56	98.95±1.02	99.95±0.10	99.90±0.14	99.44±0.63	**99.96±0.12**
OA(%)	93.23±1.76	98.68±0.77	97.34±0.49	94.50±0.53	96.58±0.95	98.23±0.41	98.6±0.43	96.31±0.87	**99.61±0.13**
AA(%)	95.99±2.20	98.17±0.14	97.34±0.49	94.50±0.53	96.58±0.95	98.23±0.41	98.6±0.43	96.31±0.87	**99.61±0.13**
Kappa×100	92.46±1.96	99.18±0.38	98.71±0.38	97.20±0.59	98.25±0.85	99.17±0.30	99.29±0.20	98.44±0.33	**99.72±0.06**

dual-branch baseline, while case 6 added both SENBR and GASA to case 5. This combination led to stable performance improvements across all datasets. Finally, case 7, which included all three modules—SENBR, GASA, and LightPyramid—achieved the best accuracy on all datasets: 98.76% on PU, 99.61% on SA, 96.20% on HongHu, and 96.01% on HanChuan. Compared to case 6, case 7 showed a performance improvement ranging from 0.8% to 2.19% across all datasets, highlighting the effectiveness of the LightPyramid module in enhancing multiscale spatial context information. These results validate the significant contribution of each submodule to the overall performance of the PS2Mamba model.

To evaluate the role of each module in our model for HSI modeling, we conducted t-distributed stochastic neighbor embedding (t-SNE) visualization experiments on deep features, with results shown in Figs. 9–12. Using the PU, SA, HongHu, and HanChuan datasets, we analyzed feature distribution under different module combinations, focusing on separability and class boundaries. In all four datasets, the feature distribution in (a) appears disordered, with unclear class boundaries and scattered intraclass samples, indicating that the model without enhancements struggles to form distinct clusters. After adding the GASA module, class boundaries become clearer, and some categories form compact clusters, particularly in the SA and PU datasets, demonstrating GASA's effectiveness. Further integration with the SENBR module improves clustering, making categories more concentrated and boundaries sharper, especially in complex scenes like HH and HC. This shows that SENBR's spectral enhancement

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TABLE III
QUANTITATIVE RESULT (ACC% ± STD%) OF HONGHU DATASET

Methods	spectralFormer TGRS 2021	CLOLN TGRS2024	AMGCFN TGRS2023	morphFormer TGRS2023	SSFTT TGRS2022	GSCVIT TGRS2024	mambahsi TGRS2024	mambaLG TGRS2025	PS2Mamba Ours
1	97.50±1.26	95.36±5.56	95.56±0.34	92.64±0.86	94.66±1.33	**98.16±1.08**	96.47±1.50	92.91±0.68	97.42±0.98
2	94.99±4.37	78.60±6.19	**99.11±0.22**	92.21±1.16	91.09±0.70	86.90±9.09	95.43±2.33	90.81±3.69	97.49±1.66
3	86.09±4.51	**98.22±1.32**	84.67±1.43	81.32±1.53	90.69±3.40	79.45±1.81	94.66±1.05	84.59±0.94	93.51±1.95
4	94.18±3.57	**99..48±0.37**	98.79±0.84	97.09±2.78	93.10±2.81	99.57±0.14	96.96±1.79	96.20±1.51	97.79±1.29
5	96.68±2.78	41.33±18.92	**99.35±0.45**	92.77±1.31	87.13±0.63	37.03±9.33	96.20±1.77	93.47±2.21	98.66±1.05
6	92.24±1.93	96.09±1.67	97.12±0.34	91.81±3.95	94.20±0.82	95.72±1.83	93.90±2.47	94.07±0.69	**98.18±1.38**
7	81.36±3.21	**92.72±1.96**	76.97±2.35	64.15±4.86	73.01±0.95	90.38±3.66	86.60±3.30	72.65±2.09	89.15±2.64
8	80.30±2.31	46.44±6.02	92.31±3.53	69.28±4.29	93.58±4.69	32.12±4.83	96.08±2.90	77.98±0.87	**99.08±0.98**
9	97.64±1.71	97.04±2.05	96.81±1.45	95.30±0.76	**99.02±2.67**	97.06±2.73	96.08±2.91	93.65±1.35	98.27±0.73
10	74.88±7.08	84.88±3.47	92.54±0.88	82.34±3.90	83.46±0.81	79.23±7.53	90.45±2.72	89.39±2.68	**95.52±1.26**
11	83.33±5.48	72.03±8.35	93.36±1.97	73.45±1.09	79.97±5.34	72.41±8.95	91.96±2.87	77.11±7.80	**95.70±1.25**
12	82.78±4.37	66.63±8.82	80.28±1.65	76.22±1.42	78.80±0.94	65.61±8.66	87.40±5.22	79.92±5.05	**90.43±4.78**
13	79.48±4.52	82.67±5.83	85.46±1.25	71.82±4.53	**94.14±1.44**	77.49±8.47	89.12±3.12	68.96±4.63	89.87±3.12
14	91.91±6.77	87.59±7.03	94.96±0.89	93.50±2.59	94.25±1.25	86.26±11.93	94.01±1.65	88.92±4.31	**95.57±2.04**
15	97.62±4.25	75.88±12.93	97.55±0.60	99.48±0.59	93.92±5.97	63.62±20.25	98.84±1.17	97.33±1.49	**99.77±0.25**
16	98.20±2.55	97.39±1.73	97.43±1.64	97.37±2.78	**98.95±0.97**	97.42±1.66	96.83±1.60	93.56±8.44	97.27±2.02
17	92.32±4.87	80.09±9.70	97.29±0.97	91.87±0.91	**99.46±1.58**	84.92±6.09	98.60±1.49	95.17±7.11	95.26±5.58
18	98.40±3.59	65.55±15.89	**98.81±1.38**	91.05±2.07	87.94±0.56	67.44±10.50	96.72±1.82	98.71±1.89	97.51±0.72
19	92.10±5.12	82.53±7.23	85.30±3.29	89.50±0.70	85.59±0.27	85.10±8.11	91.30±3.07	91.51±2.57	**94.12±1.18**
20	97.17±5.53	81.38±6.60	98.96±1.03	96.99±0.74	94.05±1.51	71.64±10.46	**99.18±0.42**	92.62±3.14	98.54±0.41
21	97.77±0.92	30.19±12.06	99.97±2.33	81.20±1.84	100±0	25.92±7.91	99.91±0.18	98.61±3.67	**100±0**
22	98.32±1.27	55.02±10.89	99.07±1.79	92.16±0.95	96.38±4.03	46.26±11.15	**99.54±0.51**	97.86±3.68	99.53±0.44
OA(%)	90.87±1.27	87.93±2.11	95.31±2.54	89.40±3.49	91.97±2.44	85.55±1.86	94.58±1.01	90.56±3.60	**96.20±0.62**
AA(%)	91.15±0.44	77.60±2.25	93.89±0.64	86.98±3.03	91.2±0.61	83.21±1.64	94.77±0.61	89.71±3.00	**96.30±0.31**
Kappa×100	88.58±1.28	85.06±2.49	92.83±1.94	86.71±2.41	89.9±2.55	82.07±2.16	94.00±1.88	88.17±7.11	**97.33±0.98**

TABLE IV
QUANTITATIVE RESULT (ACC% ± STD%) OF HANCHUAN DATASET

Methods	spectralFormer TGRS 2021	CLOLN TGRS2024	AMGCFN TGRS2023	morphFormer TGRS2023	SSFTT TGRS2022	GSCVIT TGRS2024	mambahsi TGRS2024	mambaLG TGRS2025	PS2Mamba Ours
1	87.79±1.24	**95.97±1.87**	95.46 ± 2.25	92.88±2.22	89.37±2.55	95.26±1.50	89.72±5.75	75.75±2.45	94.73±2.58
2	65.21±7.79	92.48±2.56	91.03 ± 2.68	85.93±0.93	82.53±3.11	91.22±3.05	81.37±5.69	82.94±5.18	**93.73±0.9**
3	95.08±4.49	80.03±9.18	96.17 ± 3.06	80.97±1.88	83.66±6.89	78.37±7.47	94.88±3.73	82.64±2.73	**97.4±1.57**
4	97.42±9.57	91.12±7.72	**98.75 ± 1.22**	95.24±1.70	96.91±3.06	85.17±12.38	98.46±1.35	95.98±0.85	96.98±2.81
5	99.52±0.77	56.82±12.89	99.47 ± 0.74	98.71±1.79	99.65±1.19	46.57±13.03	99.54±0.85	97.18±0.72	**99.84±0.14**
6	68.67±2.00	38.07±8.70	**96.06 ± 1.42**	62.93±1.58	79.68±2.31	35.97±7.01	76.99±4.95	77.04±4.76	93.8±1.18
7	86.51±2.30	68.87±17.46	98.15 ± 0.50	91.86±1.05	91.84±4.03	66.90±15.67	92.61±4.63	93.94±1.51	**98.35±0.57**
8	72.38±5.51	81.52±7.87	90.43 ± 3.75	66.65±3.66	76.81±3.58	82.39±5.23	70.73±7.10	78.71±2.08	**92.20±3.14**
9	65.66±7.71	68.59±6.61	90.84 ± 4.01	66.00±2.75	85.93±0.71	60.64±17.62	89.01±3.62	73.85±5.75	**96.00±2.55**
10	96.54±11.85	90.79±14.71	96.58 ± 2.23	90.55±0.88	96.21±3.99	93.22±4.31	94.44±2.18	**98.43±0.67**	96.71±0.87
11	92.90±1.90	87.86±4.63	96.59 ± 1.84	91.25±3.64	94.92±3.28	82.13±7.96	89.56±4.91	92.97±2.08	**97.85±1.32**
12	92.53±2.57	63.48±14.80	**99.51 ± 0.41**	88.62±2.21	94.46±4.23	51.92±6.01	88.83±6.91	95.64±0.70	97.94±2.07
13	79.00±3.66	50.21±6.71	81.90 ± 3.16	60.31±2.56	75.22±5.62	53.28±10.88	78.69±3.73	63.5±1.72	**83.84±3.38**
14	71.88±8.56	90.78±2.74	80.61 ± 4.55	72.82±1.78	85.89±8.54	90.64±3.07	94.42±2.41	75.19±8.19	**95.07±1.85**
15	97.95±10.97	62.42±10.67	96.37 ± 1.0	96.46±0.29	93.49±0.54	70.05±12.35	94.42±2.41	95.21±3.08	**97.06±1.58**
16	98.95±0.06	98.97±1.02	98.64 ± 1.41	98.92±0.26	96.21±4.72	**99.77±0.23**	98.28±1.62	94.18±2.92	99.08±0.49
OA(%)	85.47±1.35	85.14±2.93	95.12 ± 0.30	87.43±0.90	89.61±1.79	83.14±3.96	90.21±1.67	85.67±1.24	**96.01±0.52**
AA(%)	85.23±1.46	76.13±3.26	95.08 ± 0.26	83.88±1.66	88.92±0.83	73.97±2.06	89.20±1.36	84.94±1.02	**95.66±0.27**
Kappa×100	83.14±1.49	82.75±3.31	94.26 ± 0.39	85.34±0.49	87.92±1.35	80.57±4.33	89.07±2.31	83.38±1.40	**96.20±0.98**

suppresses redundant channels and strengthens informative ones. Finally, after integrating LightPyramid, all datasets show the most compact intraclass aggregation and clearest interclass separation, highlighting the multiscale spatial structure enhancement's impact on fine-grained pattern recognition. This module improves the feature space structure by introducing multi-receptive field context, boosting overall expressive ability.

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TABLE V
IMPACT OF DIFFERENT MODULES ON OA FOR FOUR DATASETS

Case	Component					DataSets			
	SpeBaseline	SpaBaseline	SENBR	GASA	LightPyramid	PU	SA	HongHu	HanChuan
1	√	—	—	—	—	93.65	91.58	86.00	83.58
2	√	—	√	—	—	97.64	97.61	93.35	93.39
3	—	√	—	—	—	89.08	96.27	89.11	89.02
4	—	√	—	√	—	95.57	98.21	92.18	91.90
5	√	√	—	—	—	94.07	97.86	89.24	80.12
6	√	√	√	√	—	97.95	98.87	94.01	93.97
7	√	√	√	√	√	**98.76**	**99.61**	**96.20**	**96.01**

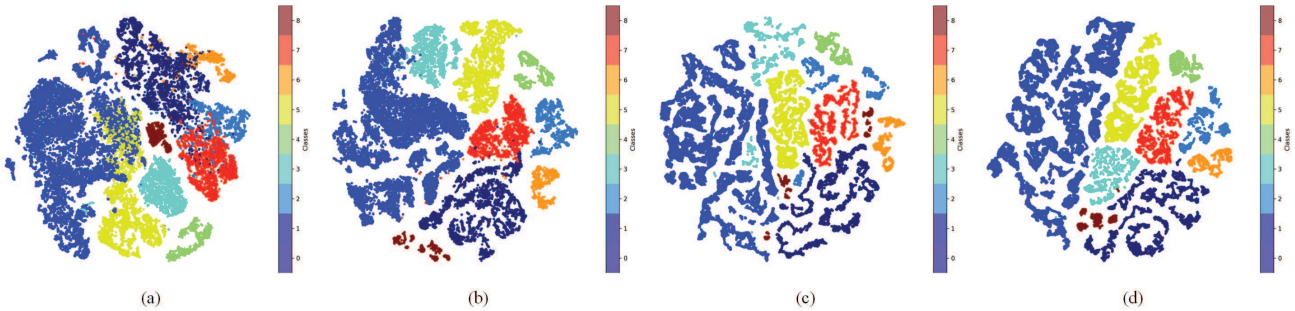


Fig. 9. t-SNE visualization results on PU Dataset. (a)–(d) t-SNE visualization results for Baseline, Baseline+GASA, Baseline+GASA+SENBR, and PS2Mamba, respectively.

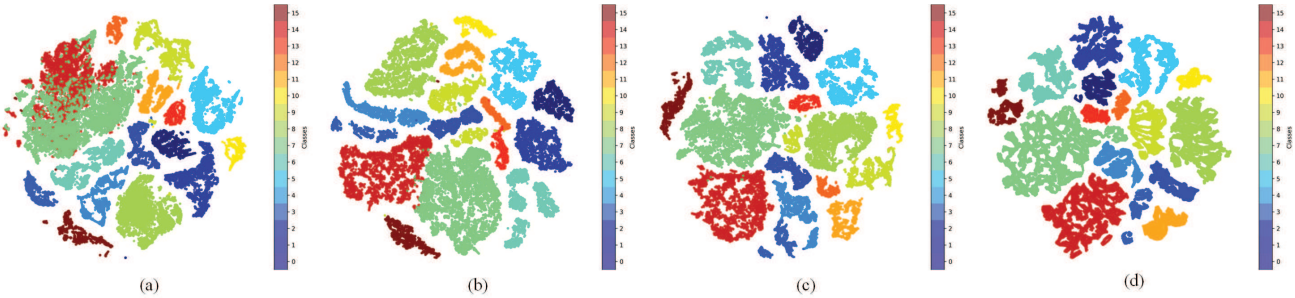


Fig. 10. t-SNE visualization results on SA Dataset. (a)–(d) t-SNE visualization results for Baseline, Baseline+GASA, Baseline+GASA+SENBR, and PS2Mamba, respectively.

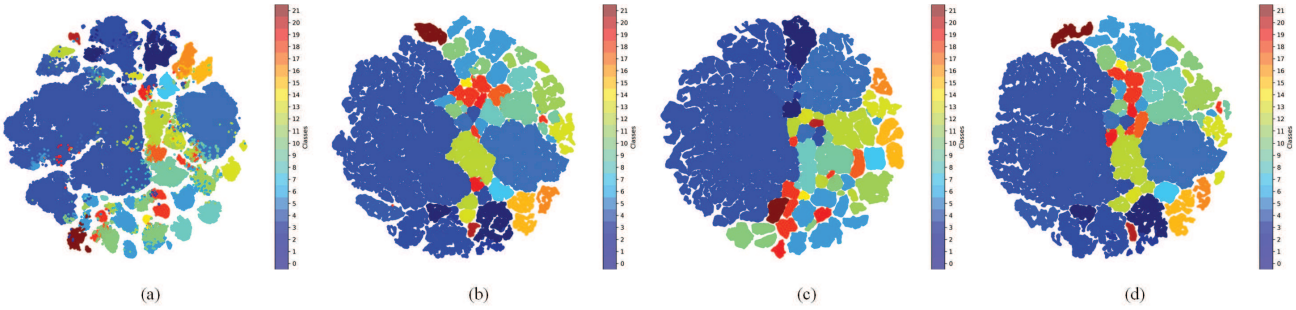


Fig. 11. t-SNE visualization results on HongHu Dataset. (a)–(d) t-SNE visualization results for Baseline, Baseline+GASA, Baseline+GASA+SENBR, and PS2Mamba, respectively.

E. Parameter Sensitivity Analysis

During the training phase, different learning rates have a significant impact on the model's classification performance. To select the optimal learning rate for the PS2Mamba method proposed in this article and to analyze the effect of pixel-level spatial features, we used OA as the evaluation metric. We conducted experiments on four datasets with different learning rates and trained PS2Mamba using patch features

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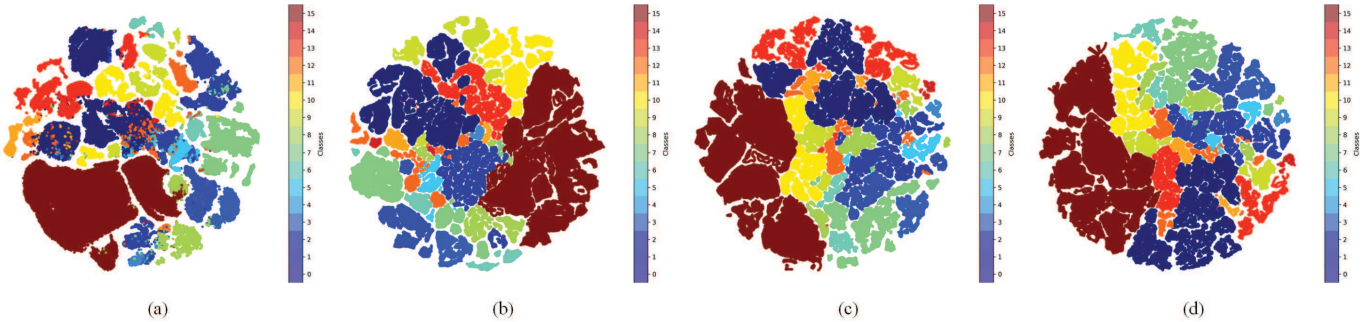


Fig. 12. t-SNE visualization results on HanChuan Dataset. (a)–(d) t-SNE visualization results for Baseline, Baseline+GASA, Baseline+GASA+SENBR, and PS2Mamba, respectively.

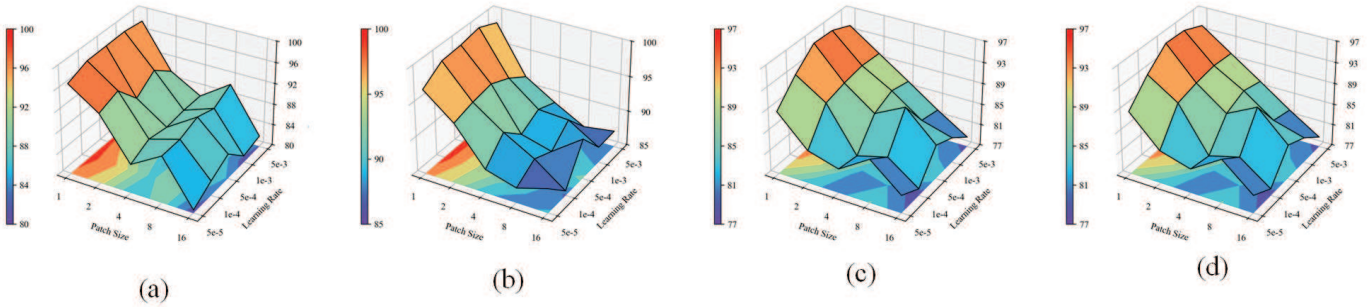


Fig. 13. Effect of different patch sizes and learning rates on OA across different datasets. (a)–(d) PU dataset, SA dataset, HongHu dataset, and HanChuan dataset, respectively.

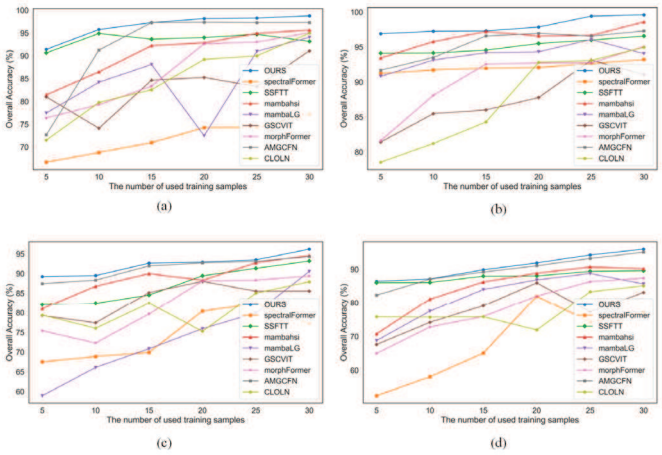


Fig. 14. Impact of different methods using different training sample sizes on the classification accuracy OA value. (a)–(d) PU dataset, SA dataset, HongHu dataset, and HanChuan dataset, respectively.

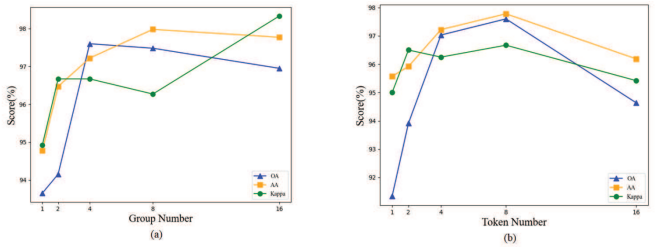


Fig. 15. Hyperparameter analysis on the PU dataset. (a) Impact of Group-Number on OA. (b) Impact of TokenNumber on OA.

of different sizes to replace pixel-level features, comparing them with pixel-level features. The selected learning rates were $\{5e^{-5}, 1e^{-4}, 5e^{-4}.1e^{-3}, 5e^{-3}\}$, and the experimental results are shown in Fig. 13.

From the results, it can be seen that on the UAV-acquired dual-high datasets, HongHu and HanChuan, as the learning rate increases, the OA first increases and then decreases. However, on the classic datasets, PU and SA, the OA first increases, then decreases, and then increases again. Therefore, we selected $3e^{-3}$ as the learning rate during the model training phase, ensuring good results on all four datasets.

Furthermore, we investigated input granularity using patch sizes $\{1, 2, 4, 8, 16\}$, where a size of 1 denotes pixel-level input. Experiments consistently demonstrated that patch-based features underperformed significantly compared to pixel-level features under identical settings. This result underscores the critical role of pixel-level features for optimal accuracy, suggesting that the fine-grained details they preserve are better aligned with the Mamba architecture's modeling strengths.

F. Impact of the Number of Training Samples

The cost of annotating HSI data is high, and thus, the training samples for HSIs are usually limited. In the experiments mentioned above, the proposed PS2Mamba method's excellent performance under small sample sizes has been demonstrated through extensive experiments. In this section, the

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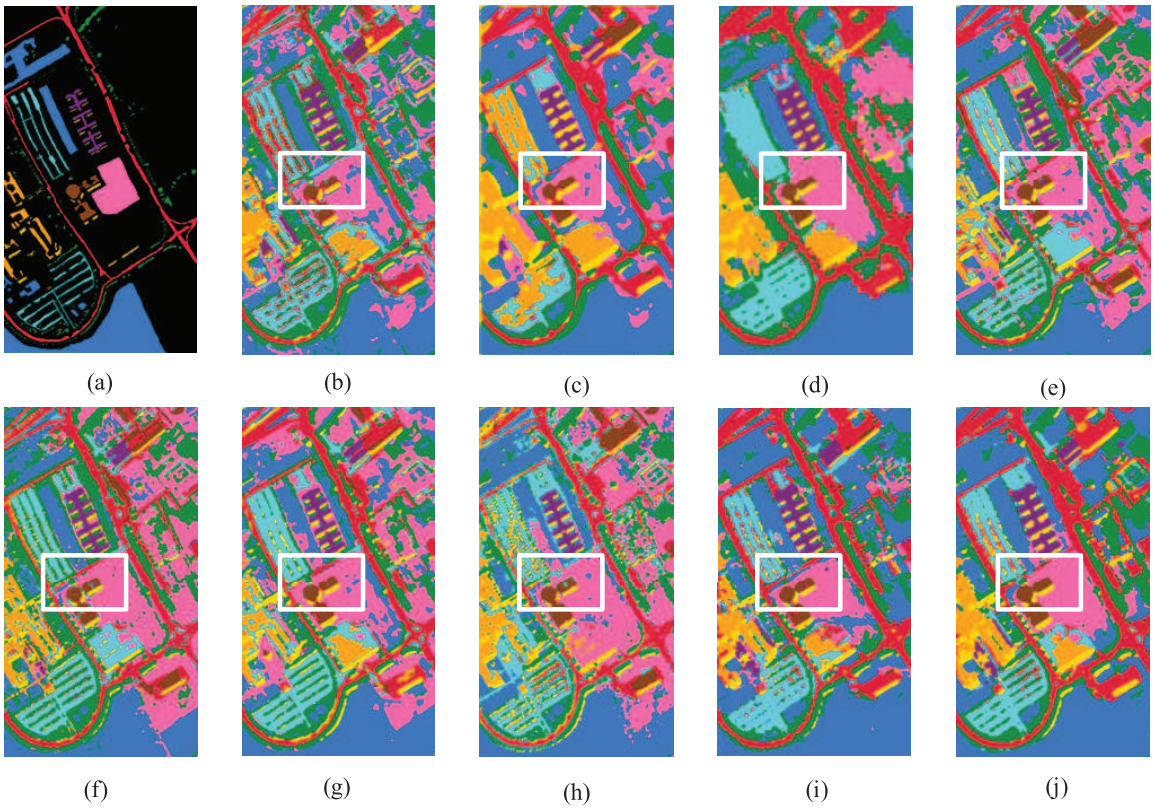


Fig. 16. Classification result graphs obtained by different methods on the PU dataset. (a) Ground truth map. (b) SpectralFormer. (c) CLOLN. (d) AMGCFN. (e) MorphFormer. (f) SSFTT. (g) GSC-Vit. (h) mambaLG. (i) MambaHSI. (j) Proposed PS2Mamba.

classification performance of PS2Mamba and all comparison methods is verified across three datasets at different training sample ratios. The number of training samples for the four datasets is selected as $\{5, 10, 15, 20, 25, 30\}$, and the impact of different numbers of training samples on OA is shown in Fig. 14, where OA represents the average of ten different random seeds. Fig. 14(a)–(c) shows the impact of different training sample sizes on OA for the PU dataset, the impact for the SA dataset, the impact for the HongHu dataset, and the impact for the HanChuan dataset, respectively.

G. Hyperparameter Analysis

To quantitatively evaluate the contributions of the proposed group modeling and local enhancement mechanisms, a hyperparameter analysis was conducted. The key parameters under investigation are GroupNumber and TokenNumber. A baseline configuration where GroupNumber = 1 and TokenNumber = 1 signifies the absence of both mechanisms, effectively reducing the model to a simpler form for ablation.

The experimental results, as summarized in Fig. 15, unequivocally demonstrate the performance gains attributable to our proposed components. As illustrated in Fig. 15(a), increasing the GroupNumber from the baseline of 1 leads to a substantial improvement in OA. This validates the effectiveness of group modeling, which enriches feature representation by facilitating diverse and specialized feature extractions from different spectral groupings. The performance is observed to saturate beyond an optimal GroupNumber, suggesting a balance between the feature diversity and model complexity.

Similarly, Fig. 15(b) reveals the impact of the TokenNumber. Introducing the local enhancement mechanism (TokenNumber > 1) yields a significant performance boost compared with the baseline. This improvement underscores the mechanism's efficacy in capturing fine-grained spatial contexts within the flattened sequence, which is crucial for precise pixel-level classification. The results indicate that there exists an optimal range for TokenNumber; an excessively high value may lead to over-segmentation of the spatial context, while a value that is too low fails to adequately leverage local information.

Consequently, the optimal values for GroupNumber and TokenNumber were selected from the peaks of the respective curves in Fig. 15, ensuring that both mechanisms operate at their highest efficacy to achieve the reported state-of-the-art performance.

H. Model Complexity and Efficiency Analysis

To evaluate the computational efficiency of the proposed method, we compare the training time, testing time, number of parameters, and floating point operations (FLOPs) with several state-of-the-art HSI classification methods on the University of Pavia dataset. The comparative results are summarized in Table VI.

As demonstrated in the table, our method achieves the fastest testing time among all competing approaches, exhibiting a remarkable reduction in inference latency compared

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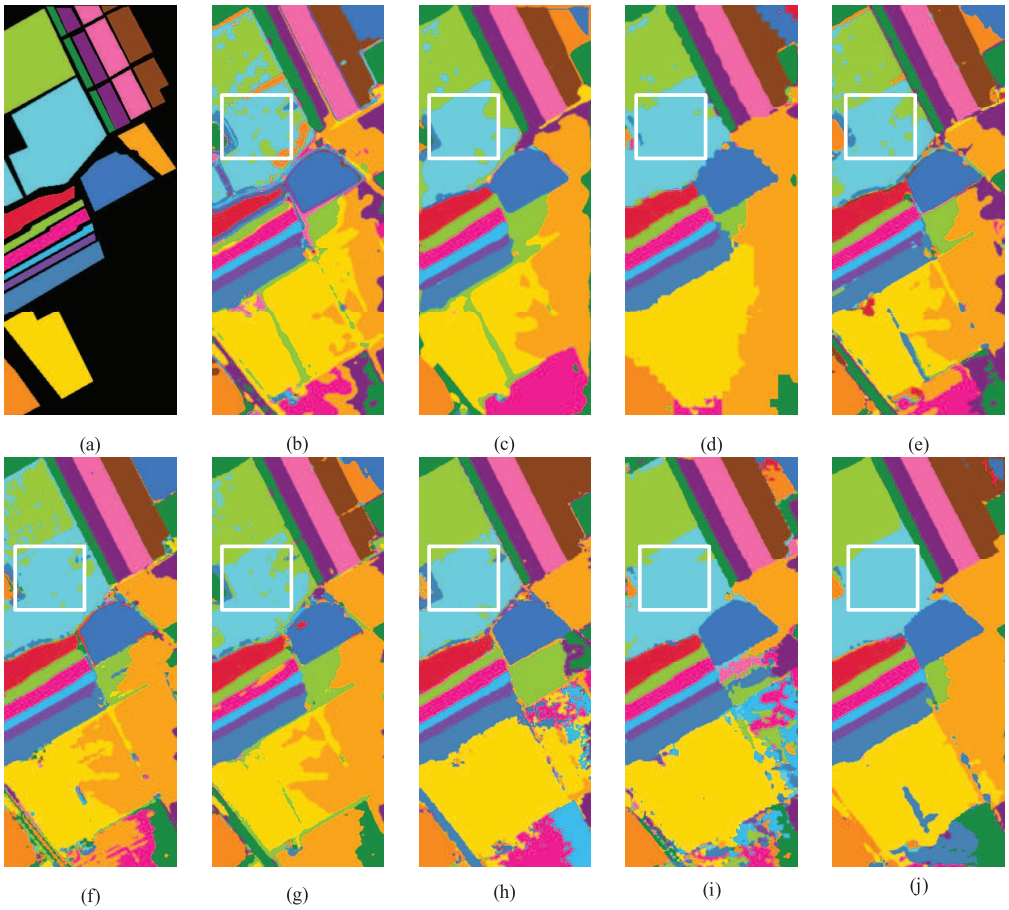


Fig. 17. Classification result graphs obtained by different methods on the SA dataset. (a) Ground-truth map. (b) SpectralFormer. (c) CLOLN. (d) AMGCFN. (e) MorphFormer. (f) SSFTT. (g) GSC-Vit. (h) mambaLG. (i) MambaHSI. (j) Proposed PS2Mamba.

TABLE VI
COMPARISON OF MODEL PARAMETERS, COMPUTATIONAL COMPLEXITY, AND RUNNING TIME OF DIFFERENT METHODS ON THE PU DATASET

Methods	spectralFormer TGRS 2021	CLOLN TGRS2024	AMGCFN TGRS2023	morphFormer TGRS2023	SSFTT TGRS2022	GSCVIT TGRS2024	mambahsi TGRS2024	mambaLG TGRS2025	PS2Mamba Ours
Train Time(s)	94.21	38.41	79.4	41.37	**11.7**	62.21	278.15	60.21	203.55
Test Time(s)	62.44	46.46	58.99	27.58	5.9	49.44	0.03	60.21	**0.03**
Parms(K)	17.73	**5.17**	320.41	223.95	148.49	153.88	412.34	287.31	412.19
FLOPS(G)	9086.19	124.44	14.98	**0.02**	50624.27	2026.3	39.18	27.77	36.41

to typical patch-based methods. This significant efficiency improvement is directly attributable to our image-level processing paradigm, which effectively eliminates the redundant computations inherent in patch-based frameworks when processing full images. In contrast, conventional patch-based methods suffer from substantial computational overlap between adjacent patches, resulting in considerably prolonged inference times.

While our model employs a relatively larger number of parameters and computational operations compared with certain lightweight architectures, it consistently delivers superior classification performance across all key metrics, including OA, AA, and kappa coefficient, as established in Section III-C. This indicates that the additional model capacity is strategically utilized to enhance the feature representation capability and ultimately improve the classification accuracy.

Furthermore, the training time required by our method remains competitive, particularly when evaluated in the context of its significant performance advantages.

In summary, our method establishes a favorable tradeoff between computational complexity and classification performance. The increased model capacity is effectively translated into exceptional classification accuracy and groundbreaking inference speed, underscoring its significant value for practical deployment, where efficiency is paramount.

I. Comparison of Classification Maps

To provide a more intuitive analysis of the classification performance of the proposed PS2Mamba method on different remote sensing hyperspectral datasets, we conducted a visualization analysis of the classification results for all comparison methods. Figs. 16–19 show the classification results

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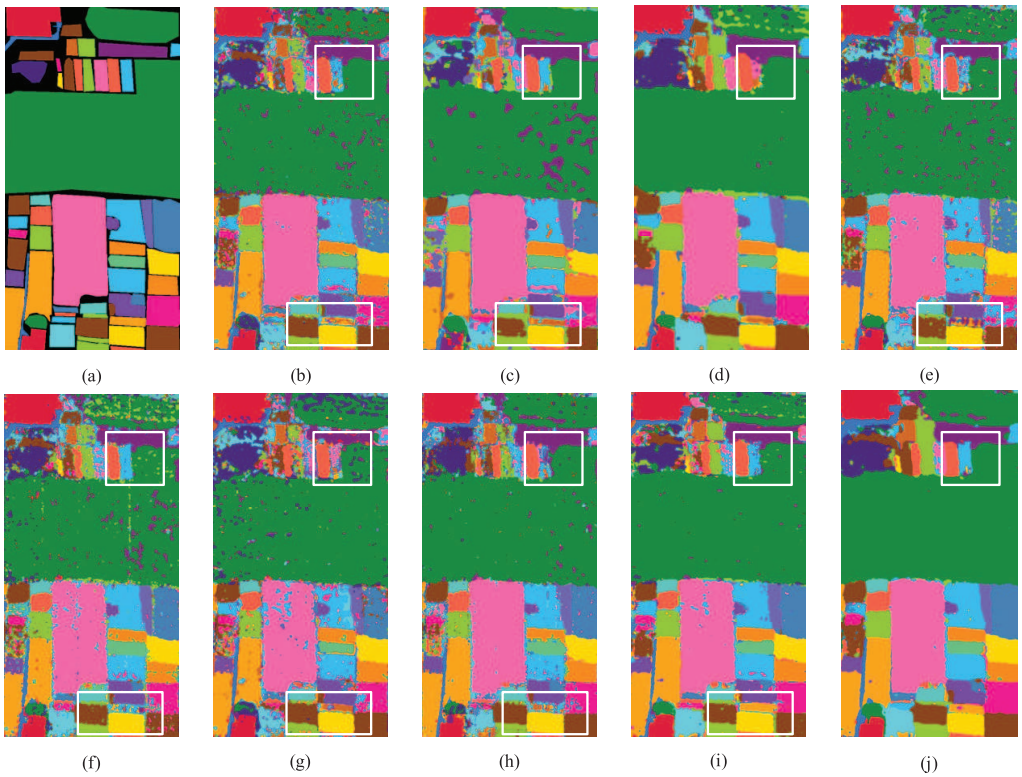


Fig. 18. Classification result graphs obtained by different methods on the HongHu dataset. (a) Ground-truth map. (b) SpectralFormer. (c) CLOLN. (d) AMGCFN. (e) MorphFormer. (f) SSFTT. (g) GSC-Vit. (h) mambaLG. (i) MambaHSI. (j) Proposed PS2Mamba.

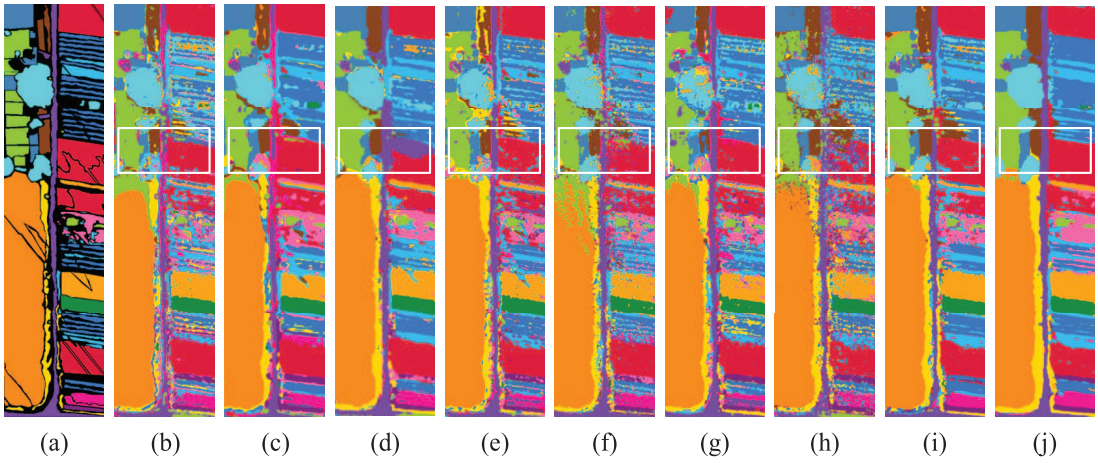


Fig. 19. Classification result graphs obtained by different methods on the HanChuan dataset. (a) Ground-truth map. (b) SpectralFormer. (c) CLOLN. (d) AMGCFN. (e) MorphFormer. (f) SSFTT. (g) GSC-Vit. (h) mambaLG. (i) MambaHSI. (j) Proposed PS2Mamba.

for each method on PU, SA, Honghu, and Hanchuan datasets, respectively. From these figures, it can be seen that, compared with other methods, the classification images obtained by PS2Mamba demonstrate superior boundary clarity, category coherence, and OA, making them more aligned with the true land cover distribution. This further validates the superiority of the method in HSI classification tasks.

IV. Conclusion

This article proposes a novel modeling framework for pixel-level HSI classification tasks, called PS2Mamba. The model addresses the core challenge of "spectral–spatial collaborative modeling" in HSIs, integrating statistical-driven spectral channel recalibration, GASA mechanisms, and a deep integration strategy of directional enhancement and state-space modeling. This comprehensive approach significantly improves the discriminative power and generalization ability of hyperspectral classification.

First, three statistical indicators—channel correlation, channel variability, and channel importance—are proposed and incorporated into the SENBR module, achieving group-based enhancement and global guidance of spectral features, thereby improving the rationality and dynamism of channel selection. Next, we design a GASA module that can adaptively select

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the optimal scale based on the global perceptual information of the image structure and introduces a multidirectional modeling mechanism to enhance the perception of different spatial structures. Finally, the proposed LightPyramid module effectively integrates multilevel spatial semantics, further enhancing the model's adaptability to complex scenes.

Systematic experiments and ablation studies were conducted on several standard hyperspectral datasets, including the UAV-acquired HongHu and HanChuan datasets, as well as the classic PU and SA datasets. The results show that PS2Mamba significantly outperforms existing HSI classification methods in the overall classification accuracy. Specifically, PS2Mamba achieved accuracies of 98.77%, 99.61%, 96.20%, and 96.01% on the four datasets, respectively. Compared with existing CNN, GCN, Transformer, and Mamba-based models, PS2Mamba provides up to a 12.91% improvement in accuracy. These results demonstrate the proposed method's outstanding robustness and adaptability, especially in complex scenes with significant land cover scale differences and under small-sample conditions.

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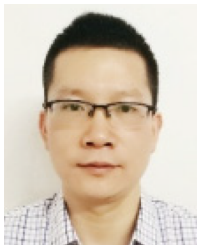
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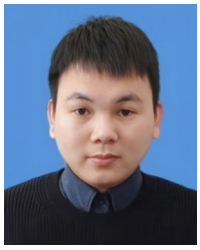
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